Responsible use of Large Language Models for Research: Using LLMs in study design

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Course website: https://bma-vandijk.github.io/RULLMR/



Programme

1. Update LUMC policy GenAI & LLM Community
2. Refresher quiz and recap
3. Homework task: online and local approaches

Learning goals

After this session you should be able to:

1. Explain key concepts and terms pertaining to the design and use of LLMs
2. Design and execute a study design task for both a cloud and local LLM and a set of papers

1. Update LUMC policy GenAI & LLM community

1.1 Update GenAI policy & LLM community

- Update w.r.t. our earlier mentions of the new GenAI policy (also available in English on Zenya):
  - Not only supportive use (preprocessing, ideation, cleaning, coding) but also everything related to research is possible with Copilot provided you are allowed to process the data
  - Responsibility regarding privacy, security, and IP sits on the side of the researcher (but Privacy and AI team are happy to advise you)

- Research-specific policy follows next month



1.2 Update GenAI policy & LLM community

- To always stay up to date w.r.t. everything LLM in the LUMC, join us!



LUMC LLM Community mailing list

2. Quiz & recap

2.1

What tasks are most suited to a LLM hosted on your own LUMC device?

- **A:** A broad scan of a research field to quickly get up-to-date
- **B:** A focused extraction task from (sensitive) free text data
- **C:** Checking references of a paper you just wrote
- **D:** Reviewing a paper presenting a groundbreaking new idea

2.2

Which point best summarizes the key issue for using LLMs with sensitive data in the LUMC?

- **A:** Whether I use the model in the cloud or locally
- **B:** Whether the model I use is biased or not
- **C:** Whether the model I use is capable hence large enough
- **D:** Whether I am entitled to share the sensitive data I want to study

2.3

What is the best description of the temperature setting on a LLM?

- **A:** Making the model most performant on the task
- **B:** Controlling the creativity of the LLM
- **C:** Ensuring the model output does not repeat itself
- **D:** Controlling the shape of the output probability distribution

2.4

What is the best description of LLM context?

- **A:** The LLM parameters/weights that store factual and linguistic information
- **B:** The prompt the user provides
- **C:** Everything that is appended to the input of a model including the prompt, chat history, tool descriptions, etc.
- **D:** The examples I provide in my prompt to ensure the model understands the task

2.5

Which statements are true?

- **A:** A LLM, in an agentic framework, can execute tasks
- **B:** A agentic LLM is still a next token predictor
- **C:** Agentic LLMs themselves define the tools they need for doing additional functions like executing code and surfing on the web
- **D:** With Ollama (or LMstudio) I can use my local LLM as agent out of the box

2.6

When an LLM generates a response, what is it doing?

- **A.** Retrieving a stored fact and returning the matching piece of evidence
- **B.** Predicting the next token from learned patterns, conditioned on the current context
- **C.** Checking each sentence against many databases before showing it
- **D.** Proving the claim with logic, then translating the proof into prose

2.7

What is context rot?

- **A.** Training data becoming quickly outdated after a model is released
- **B.** A setting to prevent your chat history from becoming too long
- **C.** Information inside a long input losing its influence on the next prediction
- **D.** The quality drop when a model is downloaded for local use

2.8

A hallucinated LLM output can sound as fluent as a true one. Why?

- **A.** Because the LLM knows about the probabilities of its next word predictions and if these are high, the model will sound confident
- **B.** Because there is no difference between false and true continuations from the perspective of the model
- **C.** Because models are after pre-training, post-trained to be helpful rather than correct
- **D.** Because all LLMs reason and this boosts their fluency

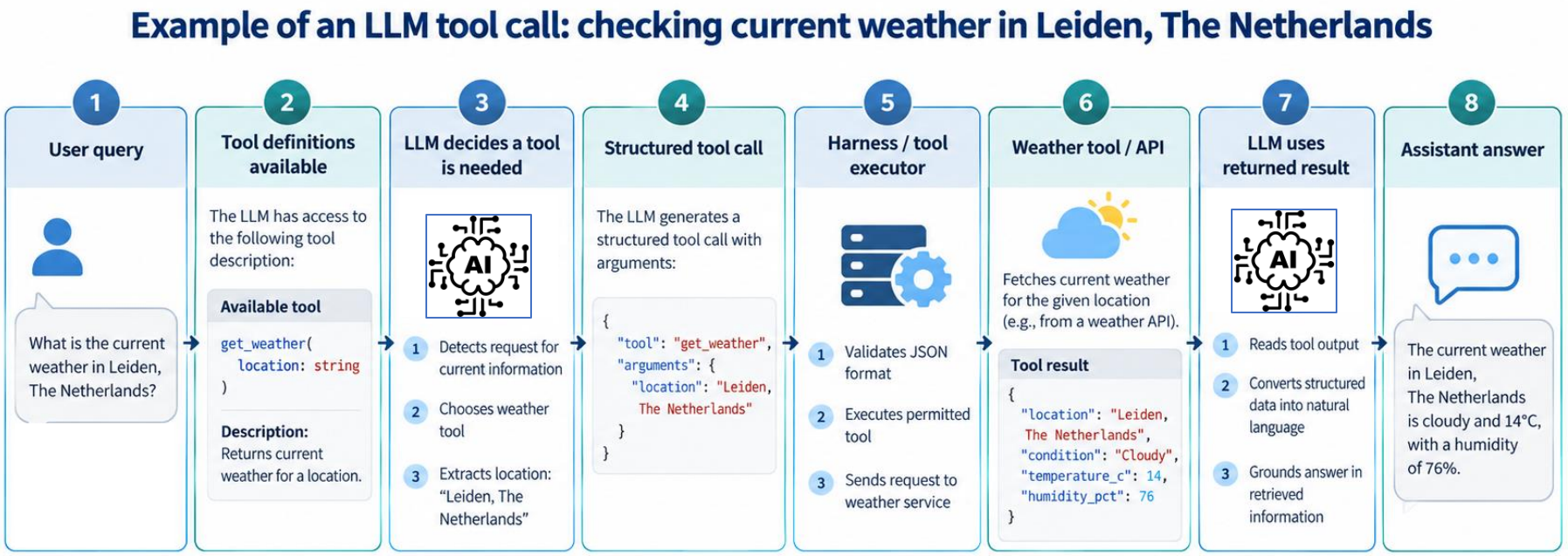
2.9

Select the considerations unique to working with LLMs:

- **A.** Whether I really need AI or this would be overkill
- **B.** How much energy and water I want to consume
- **C.** To what extent the model I choose or train is biased
- **D.** Whether the cloud platform I use can be trusted with my data

2. (Recap LLMs and agents)

2.10 (Recap LLMs and Agents)



- Tools can also concern using a calculator, executing Python, running (power)shell commands, file reading, file writing, calling APIs, creating a subagent, etc.

3. Homework task: a local approach

3.1 The task

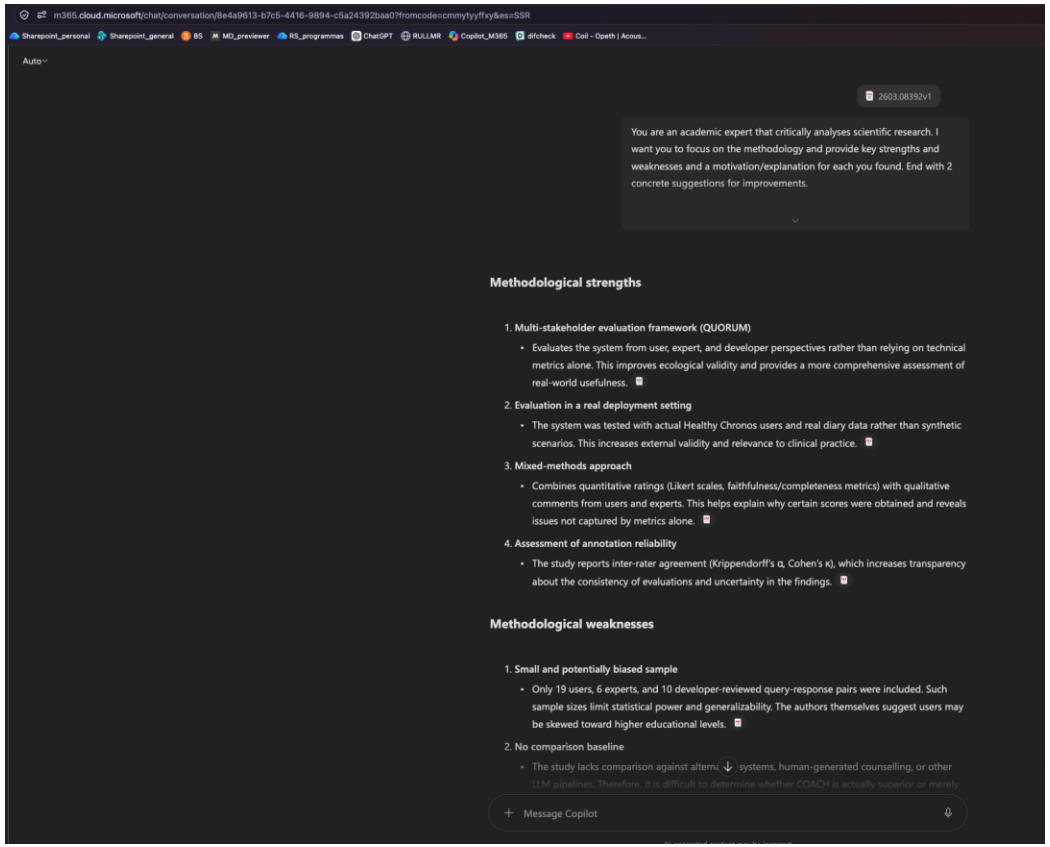
Remember from the homework assignment PDF:

- Before using an LLM, write two or three sentences about your own view of the design and what you expect the model to suggest or criticise.
- Focusing now on the 'critique and improve chosen design' task

3.2 Using a cloud model

First I use Copilot to see whether a more powerful model aligns with my expectations and to find the upper bound

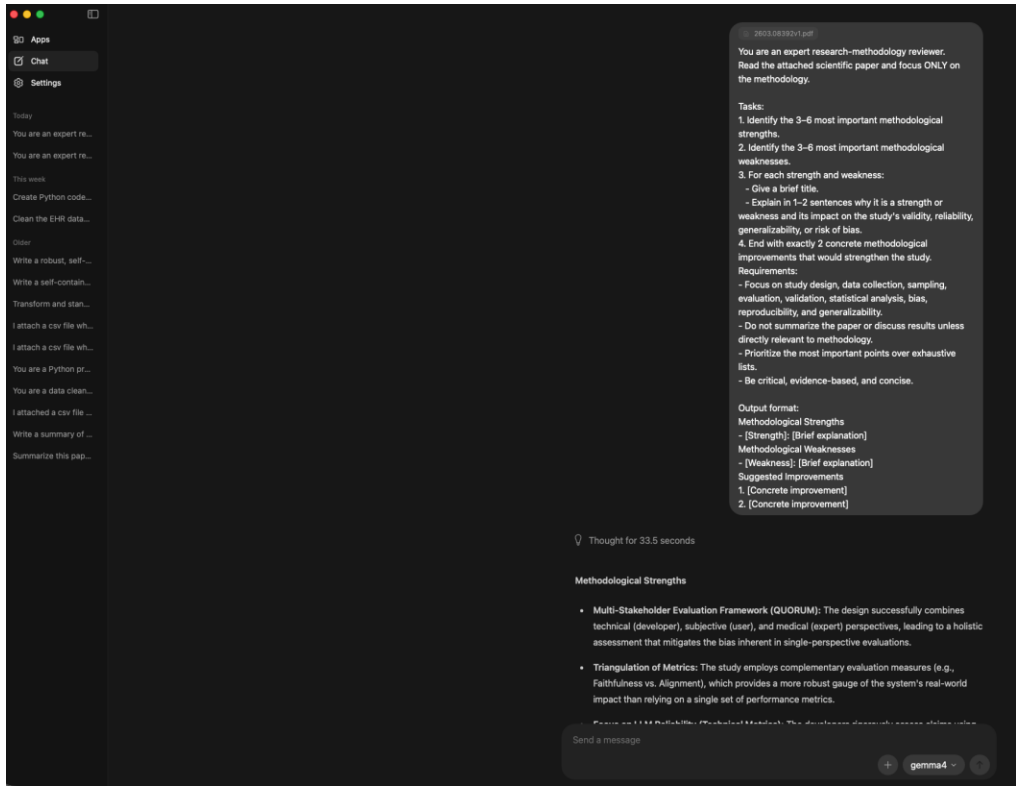
[You are an academic expert that critically analyse | Shared Copilot Chat]



3.3 Comparing quality with local model

Now I pretend this paper had a really groundbreaking experiment and result that I am not confident sharing with a third party... so I go local!

- Gemma 4 provides good results, partly overlapping and partly complementary
- But: running the same prompt + same input multiple times gives varying results



3.4 A simple workflow for fully reproducible processing

So I try to use the CLI for maximum reproducibility

[Give me a simple, step-by-step guide to extract me | Shared Copilot Chat]



Please indicate attendance

